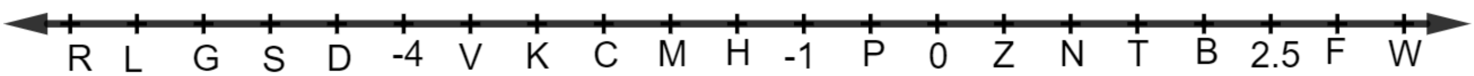
Grade 8 Name____________________
Unit 2 Assessment Date_________________
No Calculator Period__________

1. A number line is shown.



Give the letter of the point that is closest to each value.

a. $2\pi - 9$

Point	

b. $\sqrt{3}$

Point	

c. $\sqrt[3]{9} - 8$

Point	

2. A list of numbers is shown.

$$\frac{27}{4}, -3.2, -\sqrt[3]{60}, \sqrt{48}, 7.05$$

List the numbers from least to greatest.

Least to Greatest	

3. For each equation, select all of the solutions.

		Solutions				
		$x = -4$	$x = 4$	$x = -8$	$x = 8$	not a real solution
a.	$x^2 = 64$	☐	☐	☐	☐	☐
b.	$x^3 = -64$	☐	☐	☐	☐	☐
c.	$x^2 = -64$	☐	☐	☐	☐	☐

4. Compare each pair of expressions using the symbols <, >, or =.

a. $8 - \pi$ ○ < ○ > ○ = $2(2.6)$

b. $\frac{9}{4}$ ○ < ○ > ○ = $\frac{3}{4} \cdot \sqrt[3]{8}$

c. $(4)^2$ ○ < ○ > ○ = $\sqrt{256}$

d. $\sqrt{78}$ ○ < ○ > ○ = 3π

5. Complete the statements.

When x is any real number other than 0 and it is multiplied by itself, the

result will
○ always
○ sometimes
be a
○ positive
○ negative
value.

Therefore, equations in the form $x^2 = p$, where p is a whole number will

have
○ only one solution.
○ two solutions.
○ three solutions.

6. A list of numbers is shown.

$$\frac{\sqrt{12}}{2}, \qquad -0.1\sqrt{144}, \qquad -\sqrt[3]{9}, \qquad \sqrt{43}, \qquad -0.23\pi, \qquad \frac{57}{10}$$

Write each number in the appropriate column of the table based on the definitions of rational and irrational numbers.

Rational Number	**Irrational Number**